



Final Exam

Note: Answer All questions only

Q1:- Answer Ten only from the following

(10 Mark)

- 1- Consider the following code. What will be the output?

```
int[] arr = new int[5];
for (int i = 0; i < arr.length; i++) {
    arr[i] = i * i;
}
System.out.println(arr[3]);
```

0 1 2 3 4

A) 6

B) 9

C) 3

D) 0

- 2- What is the result of this code?

```
int[] nums = {10, 20, 30};
System.out.println(nums[nums.length - 1]);
```

A) 10

B) 20

C) 30

D) Error

- 3- What does the following code output?

```
int[][] matrix = {
    {1, 2, 3},
    {4, 5, 6} };
System.out.println(matrix[1][2]);
```

A) 6

B) 5

C) 2

D) 3

- 4- Given this code, what is the output?

```
public class Test {
    static int add(int a, int b) {
        return a + b;
    }
    static double add(double a, double b) {
        return a + b;
    }
    public static void main(String[] args) {
        System.out.println(add(2, 3));
    }
}
```

A) 5

B) 5.0

C) Compilation error

D) 5.

- 5- What will be printed?

```
public class Example {
    void print(int a, int... b) {
        System.out.println(b.length);
    }
    public static void main(String[] args) {
        new Example().print(1, 2, 3, 4);
    }
}
```

A) 3

B) 4

C) 2

D) Error



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6- What is the result of the following?

```
String s1 = new String("Hello");  
String s2 = "Hello";  
System.out.println(s1 == s2);
```

- A) true B) false C) Compilation error D) Hello

7- What is the output?

```
String str = "Java";  
str.concat(" Programming");  
System.out.println(str);
```

- A) Java Programming B) Java C) Compilation error D) Programming Java

8- What is the output?

```
String str = "Java Programming";  
System.out.println(str.substring(5, 8));
```

- A) Prints Pro B) Prints ram C) Prints ogr D) Prints gra

9- Which of the following is NOT a valid string method in Java?

- A) charAt() B) substring() C) equalsIgnoreCase() D) splitAt()

10- What will the following Java code output?

```
public class StringTest {  
    public static void main(String[] args) {  
        String s1 = "Java";  
        String s2 = s1.concat("Script");  
        String s3 = "JavaScript";  
        System.out.println((s2 == s3) + " " + s2.equals(s3));  
    }  
}
```

- A) true true B) false true C) true false D) false false

11- What is the output of the following?

- A) 3 B) 4 C) 5 D) Compilation error

Q2: Select the correct code for five questions only

(10 Marks)

1- Which of the following correctly declares and initializes a one-dimensional array in Java?



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- A) `int arr = new int[5]{1, 2, 3, 4, 5};` B) `int arr[] = {1, 2, 3, 4, 5};` C) `int arr[] = new int(5);`
 D) `int[] arr = (1, 2, 3, 4, 5);`

2- Which code correctly declares and accesses a 2D array element?

- A- `int[][] matrix = new int[3][3];` B- `int matrix[3][3];` C- `int[][] matrix = {{1,2},{3,4}};`
`System.out.println(matrix(1)(2));` `System.out.println(matrix[1][2]);` `System.out.println(matrix[1][1]);`
 D- `int matrix = {{1,2},{3,4}};`
`System.out.println(matrix[1][1]);`

3- Which of the following is a valid method definition in Java?

- A- void method int a, int b) B- `public void method(int a, int b) { System.out.println(a + b); }` C- `function add(int x, int y) { return x + y; }`
`{ return a + b; }`
 D- `int add(x, y) { return x + y; }`

4- Which code correctly gets the character at the 4th position of a String (index 3)?

- A- `String str = "Java";`
`System.out.println(str[3]);` B- `String str = "Java";` C- `String str = "Java";`
`System.out.println(str.charAt(3));` `System.out.println(str.get(3));`
 D- `String str = "Java";`
`System.out.println(str.at(3));`

5- Which code correctly counts the number of vowels in a string?

- A- `String s = "Programming";`
`int count = 0;`
`for (char c : s.toCharArray()) {`
`if ("aeiou".contains(c))`
`count++; }` B- `String s = "Programming";`
`int count = 0;`
`for (char c : s.toCharArray())`
`{`
`if ("aeiouAEIOU".indexOf(c)`
`!= -1) count++;`
`}` C- `String s = "Programming";`
`int count = 0;`
`for (int i = 0; i < s.length(); i++) {`
`char c = s.charAt(i);`
`if (Character.isVowel(c)) count++; }`
 D- `String s = "Programming";`
`int count = s.chars().filter(c ->`
`"aeiou".indexOf(c) > 0).count();`

6- Which of the following correctly creates a string and compares it ignoring case?

- A- `String a = "hello";`
`String b = "Hello";`
`System.out.println(a == b);` B- `String a = "hello";` C- `String a = "hello";`
`String b = "Hello";` `String b = "Hello";`
`System.out.println(a.equalsIgno` `System.out.println(a.equalsIgno`
`reCase(b));` `reCase(b));`
 D- `String a = "hello";`
`String b = "Hello";`
`System.out.println(a.compareToIgnoreCase`
`(b));0).count();`



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Q3:- Answer five points (True , False)

(10 Marks)

- 1- Java allows jagged arrays (arrays of rows with different lengths) using 2D arrays.
- 2- A multidimensional array in Java must have uniform row lengths.
- 3- Static methods can be called without creating an object of the class.
- 4- Two strings with the same content are always equal with ==.
- 5- You can use split() to divide a string using a regular expression.
- 6- str.trim() removes all white spaces from a string.

Q4: Select the right answer for five only

(10 Marks)

- 1- Tests code comprehension and logical execution flow.

```
int[] nums = {10, 20, 30};
```

```
System.out.println(nums[1] + nums[2]);
```

A) 30 B) 40 C) 50 D) Compilation error

2- Match the method with its description:

- | | |
|-------------|-------------------------------|
| A. equals() | 1. Returns string length |
| B. length() | 2. Compare reference identity |
| C. == | 3. Compare string content |

3- Which of the following are valid method declarations in Java?

- A) void test(int a) B) int test(int x, int y) C) test void(int a)
D) public void test(String s)

4- Arrange the following code lines to form a working method:

1. } 2-public static void greet() {
3- greet(); 4-System.out.println("Hello World");

5- Match each method call to its correct output:

Method Call	Output
str.length()	A. 'J'
"Java".charAt(0)	B. false
"abc".equals("ABC")	C. 4

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University of Kufa

Faculty of Computer Science & Mathematics

Department: computer science

Course Name: (Advance Java)



Test B

Academic Year :2024-2025

Course Stage/Level: 1/ Second course

Date of Exam: 11/5/2025

Time: One Hour

حيدر علي عبد الزهرة
-A-
(5 Mark)

Q1:- Answer the following

1- What is the output of the following code?

```
int[][] arr = {{1, 2, 3}, {4, 5, 6}}  
System.out.println(arr[1][2])
```

- A. 4
- B. 5
- C. 6
- D. Error

2- Which statement is true about jagged arrays in Java?

- A. All rows must have the same number of columns
- B. They are illegal in Java
- C. Rows can have different lengths
- D. You can only create them with fixed size

3- Which of the following correctly uses a professional for loop to print all elements of a 2D array?

```
int[][] arr = {{1, 2}, {3, 4}};
```

- A- for (int i : arr)
System.out.println(i);
- B- for (int[] row : arr)
for (int val: row)
System.out.println(val);
- C- for (int i = 0; i < arr.length; i++)
for (int j : arr[i])
System.out.println(j);
- D- All of the above

4- What does the substring(2, 4) method return for the string "Computer"?

- A) mp
- B) mpu
- C) put
- D) mpu

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Academic Year : 2024-2025
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Time: One Hour

Test B

5- What is the output of the following code?

```
String str = "one,two,,four";  
String[] parts = str.split(",");  
System.out.println(parts.length);
```

- A) 3
- B) 4
- C) 5
- D) Compilation Error

5)

Q2: Write java program to create array of 10 Names then passed (5 Marks)
this array to

- 1- Find (Names [], Key) to find key("Ali") and print results in method.
- 2- Sort array using SortName(Names[]) then print sorted array
- 3- Change each first char of name in array to Uppercase()
- 3- Use (Switch) to make options for call methods.

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Final examination

Q1: Multiple choices (answer ten only):

- 1- Among the following which contains scanner class
a- Java.sql.timestamp b- java.io.time c- java.util

- 2- Find the output of the following program.

```
public class Solution{
    public static void main(String[] args){
        short x = 10;
        x = x * 5;
        System.out.print(x);
    }
}
```

a- 50 b- 20 c- compiler error

- 3- Select the valid statement.

- a- char[] ch=new char(5) b- char[] ch=new char[5] c- char[] ch=new char[]

- 4- Select the valid statement to declare and initialize an array.

- a- int[] A={ } b- int[] A={1,2,3} c- int[][] A={1,2,3}

- 5- Find the value of A[1] after execution of the following program.

```
int[] A = {0,2,4,1,3};
for(int i = 0; i < A.length; i++){
    A[i] = A[(A[i] + 3) % A.length];
}
```

- a- 0 b- 1 c- 3

- 6- equals () return result

- a. Boolean b- int Value c- String

- 7- Identify the output of the following program.

```
String str = "abcde";
System.out.println(str.substring(1, 3));
```

- a. abc b- bc c- cd

- 8- Identify the output of the following program.

```
Public class Test{
    Public static void main(String args[]){
        String str1 = "one";    String str2 = "two";
        System.out.println(str1+str2);
    }
}
```



Final examination

- 9- Identify the return type of a method that does not return any value
- a- one b- onetwo c- tyoone
a. int b. double c. void

- 10- Identify the output of the following program.

```
String str = " Hellow ";
System.out.println(str.Trim());
```

- a- Hellow b- tHellow c- Hellowt

- 11- How many times will "Interviewbit" be printed.

```
Int count = 0;
do{
    System.out.println("Interviewbit");
    count++;
} while(count < 10);
```

- a- 9 b-11 c-10

(10 Marks)

Q2:

Write java program to create three method , area of circle area,of square , area of triangle , with passing parameter from keyboard, use (switch) to make options for methods

Q3: Describe two for the followings:

- 1- What is different between split() and substring() with example?
- 2- What is different between create string using String literal and new keyword with example?
- 3- What is different between method with return value and method with no return value ?

(10 Marks)

Q4: Write java program to create array Op[]={+, -, *, /, ^} then apply all these mathematic operations on variables (A, B).

Good luck

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Test A

Q1:- Answer the following

1. Which of the following correctly creates and initializes a 2D integer array in Java?

- A. `int[][] matrix = new int[2][3];`
- B. `int matrix = new int[2][3][];`
- C. `int matrix[][] = new int[[]];`
- D. `int[2][3] matrix = new int[[]];`

2. Identify the correct way to declare a jagged (irregular) 2D array:

- A. `int[][] jagged = new int[3][3];`
- B. `int[][] jagged = {{1, 2}, {3, 4, 5}};`
- C. `int[][] jagged = new int[3][3][3];`
- D. `int[][] jagged = {{1, 2}, {3, 4}};`

3. Which of the following is not a valid method for iterating over a 2D array in Java?

- A. `for (int i = 0; i < arr.length; i++) for (int j = 0; j < arr[i].length; j++)`
- B. `for (int[] row : arr) for (int val : row)`
- C. `while (i < arr.length) while (j < arr[i].length)`
- D. `for (int i : arr) for (int j : arr[i])`

4. What is the result of `text.substring(2, 5)` if `text = "Elephant"`?
0 1 2 3 4 5 6 7

- A. epht
- B. epha
- C. epha
- D. eph

5. What is the output of the following code?

```
String items = "pen,pencil,eraser";
```

```
String[] list = items.split(",");
```

```
System.out.println(list[2]);
```

- A. pen
- B. pencil
- C. eraser
- D. ,eraser



Test A

(5 Marks)

Q2: Create a class called Invoice, this class have

1- Create array items[] = { 1:sugar , 2:Tea, 3:rice }

2- Create array called prices

1	200
2	400
3	600

3- Find Maximum and minimum price in second column

4- Create method FindKey(items, prices, Key) to find key="Tea" , then print its price from prices array.

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اسم: اسد سرور جاسم
(B)

First Test

Q1: Consider the following Java program. What will be the output when it is executed?

```
public class ReverseArrayLogic {  
    public static void main(String[] args) {  
        int[] arr = {3, 1, 4, 1, 5, 9};  
        int sum = 0;  
        for (int i = 0; i < arr.length / 2; i++) {  
            int temp = arr[i];  
            arr[i] = arr[arr.length - 1 - i];  
            arr[arr.length - 1 - i] = temp;  
        }  
        for (int i = 0; i < arr.length; i++) {  
            if (i % 2 == 0) {  
                sum += arr[i] * 2;  
            } else { sum -= arr[i]; }  
        }  
        System.out.println("Sum: " + sum);  
    }  
}
```

temp
arr[i]
temp

{9, 5

output: Sum: 9

Q2: Write java program to create 2D array 3× 4 rows for students and columns for Grads of topics.

Find the failed student in the second topic.



Q1:- Answer the following

1- What is the output

```
int[] a = {1, 2, 3, 4};  
int s = 0;  
for(int i = 0; i < a.length; i += 2)  
    s += a[i];  
System.out.println(s);
```

A) 10

B) 6

C) 4

D) 5

2- Which of the following is TRUE about methods?

A) A method must always return a value

C) A method can have parameters and no return value

B) A method cannot have parameters

D) Methods are only used in main

3- Find the output

```
int[] a = {10, 20, 30};  
int temp = a[0];  
a[0] = a[2];  
a[2] = temp;
```

System.out.println(a[0] + " " + a[2]);

A) 10 30

B) 30 10

C) 20 30

D) 30 20

4- Choose the correct method

```
static void max(int[] a){  
    int m = 0;  
    for(int i=0; i<a.length; i++){  
        if(a[i] > m)  
            m = a[i];  
    }  
    return m;  
}
```

A) static int max(int[] a)
C) static double max(int a)

B) static void max()
D) static int max()

5- Choose the correct condition

```
int[] a = {3, -2, 5, -1};  
for(int i=0; i<a.length; i++){  
    if(a[i] == 0)  
        System.out.println(a[i]);  
}
```

A) if(a[i] < 0)

B) if(a[i] == -1)

C) if(a[i] == "0")

D) if(a[i] > "0")

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Test A

Academic Year :2025-2026

Course Stage/Level: 1/ Second course

Date of Exam: 10/5/2025

Time: One Hour

Q3: A) public class Q1 {

```
public static void main(String[] args) {  
    int[] a = {10, 20, 4, 45, 99};
```

```
    int result = secondLargest(a);
```

```
    System.out.println("Second Largest = " + result);  
}
```

```
// Write method here  
}
```

Write the missing method

B)

```
public class Q2 {
```

```
    public static void main(String[] args) {
```

```
        int[] a = {0, 1, 0, 3, 12};
```

```
        shiftZero(a);
```

```
        for(int i = 0; i < a.length; i++)
```

```
            System.out.print(a[i] + " ");
```

```
    }
```

```
// Write method here
```

```
}
```

Write the missing method

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(5 Marks)



Test B

Q1:- Answer the following

1- What is the output?

```
int[] a = {4, 2, 7, 2};  
int sum = 0;  
for(int i = 1; i < a.length; i++)  
    sum += a[i] - a[i-1];  
System.out.println(sum);
```

- A) 3 B) -2 C) -5 D) 0

2- Which method header means: pass parameter and return value?

- A) static void print() B) static void sum(int a)
C) static int sum(int a, int b) D) public class sum()

3- What is the main error in this code

```
int[] x = {1, 2, 3};  
System.out.println(x[3]);
```

- A) Missing semicolon B) Wrong data type
C) Array index out of bounds D) No error

4- Find the out put

```
int[] a = {3, 1, 4, 1, 5};  
int max = a[0];
```

```
for(int i = 1; i < a.length; i++){  
    if(a[i] > max)  
        max = a[i];  
}  
System.out.println(max - a[1]);
```

- A) 4 B) 5 C) 2 D) 1

5- Find Output

```
static int test(int a){  
    return a * a - 2;  
}
```

```
public static void main(String[] args){  
    int x = 4;  
    System.out.println(test(x) + test(2));  
}
```

- A) 18 B) 16 C) 20 D) 14

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Academic Year :2025-2026

Course Stage/Level: 1/ Second course

Date of Exam: 10/5/2025

Time: One Hour

Test B

Q2: A)

(5 Marks)

```
public class Q3 {  
    public static void main(String[] args) {  
        int[] a = {1, 2, 2, 3, 3, 3};  
        countFrequency(a);  
    }  
    // Write method here  
}
```

Write the missing method

B)

```
public class Q8 {  
    public static void main(String[] args) {  
        int[] a = {1, 2, 4, 5};  
  
        System.out.println(missingNumber(a));  
    }  
  
    // Write method here  
}
```

Write the missing method

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Note: Answer Four questions only

Q1:- Choose the correct answer for each of the following

(10 Mark)

1- What is the output of the following code?

```
String s = "Computer";
System.out.println(s.charAt(3));
```

A) p B) m C) u D) t

2- What is the output of the following code?

```
String s = "Java Programming";
System.out.println(s.contains("gram"));
```

A) false B) Compilation Error C) true D) 4

3- Which method removes spaces from the beginning and end of a string?

A) split() B) trim() C) substring() D) contains()

4. What is the output of the following code?

```
int[] a = {2,4,6,8};
int sum = 0;
for(int i=0;i<a.length;i++)
    sum += a[i];
System.out.println(sum/a.length);
```

A) 4 B) 5 C) 6 D) 20

5. Which method header is correct for a method that receives an integer array and returns the largest element?

A) static void max(int[] a)
C) static int max()B) static int max(int[] a)
D) static double max(int a)

Q2: Correct the Following Code Segments

(10 Marks)

1- The following code should count negative numbers in an array.

```
int[] a = {3,-2,5,-1};
int count = 0;
for(int i=0;i<=a.length;i++){
    if(a[i] < 0)
        count++;
} System.out.println(count);
```



2- The following method should return the average of array elements.

```
static void average(int[] a) {
    int sum = 0;
    for(int i=0; i<a.length; i++)
        sum += a[i];
    return sum/a.length;
}
```

3-Array Index Problem

```
int[] a = {10, 20, 30, 40};
for(int i = 0; i <= a.length; i++)
    System.out.println(a[i]);
```

4- Method Return Type Problem

```
static void findMax(int[] a){
    int max = a[0];
    for(int i = 1; i < a.length; i++)
        if(a[i] > max)
            max = a[i];
    return max;}
}
```

5- Second Largest Method

```
static int secondLargest(int[] a){
    int first = 0;
    int second = 0;
    for(int i = 0; i < a.length; i++){
        if(a[i] > first){
            second = first;
            first = a[i];
        }
    }
    return first; }
```

a = { 20, 20, 30, 40 }

Q3: For each case below, write the most suitable String method used to obtain the result. (10 Marks)

Original String	Result	Method
"IRAQ "	"IRAQ"	?
"java"	"JAVA"	?
"PROGRAMMING"	"programming"	?
"University of Kufa"	Check if "Kufa" exists	?
"Artificial Intelligence"	Extract "Intelligence"	?

Q4: A) Write the missing method

(5 Marks)

```
public class Q2A {
    public static void main(String[] args) {
        int[] a = {1, 2, 2, 3, 3, 3};
        countFrequency(a);
    }
}
```

B) Write the missing method

(5 Marks)

```
public class Q4B {
    public static void main(String[] args) {
        int[] a = {1, 2, 4, 5};
        System.out.println(missingNumber(a));
    }
}
```

Q5: Write True or False for the following statements

(10 Marks)

1. Two strings created as follows:
 String s1 = "Java";
 String s2 = new String("Java");
 The expression $s1 == s2$ returns true.

2. For the array:
 int[] a = new int[5];
 The value of a[4] is null before assigning any value.

3. The following code prints all elements of the array without causing any error:

```
for(int i = 0; i <= arr.length; i++)
    System.out.println(arr[i]);
```

4. A method declared as:
 static void test()
 can contain the statement:
 return 5;

5. The output of the following code is 5:
 String s = "Computer";
 System.out.println(s.substring(3,5).length());

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